7 **(a)** Inside the magnetic field, the proton experiences a magnetic force that is always perpendicular to its velocity. This force provides the centripetal force which changes the direction of velocity of the proton but the speed of the proton remains constant.

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Since the magnitude of the force (dependent on speed) remains constant, the proton moves along the arc of a circle while in the uniform magnetic field.

(b) **(i)** The magnetic force on the proton provides the centripetal force

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq}$$

$$= \frac{(1.67 \times 10^{-27} \times 2.0 \times 10^5)}{(3.5 \times 10^{-3})(1.60 \times 10^{-19})}$$

$$= 0.5964$$

$$= 0.60 \text{ m}$$

(ii) Time spent in the magnetic field

$$\Delta t = \frac{1}{4}T = \frac{\frac{1}{4} \times 2\pi r}{v} = \frac{\pi(0.60)}{2 \times 2.0 \times 10^5} = 4.7124 \times 10^{-6} \text{ s}$$

$$\Delta v = \sqrt{v_i^{\,2} + v_f^{\,2}} = \sqrt{2}v$$

$$\langle F \rangle = \frac{\Delta p}{\Delta t} = \frac{m\Delta v}{\Delta t}$$

$$= \frac{(1.67 \times 10^{-27})\sqrt{2}(2.0 \times 10^5)}{4.7124 \times 10^{-6}}$$

$$= 1.002310^{-16}$$

$$= 1.0 \times 10^{-16} \text{ N}$$